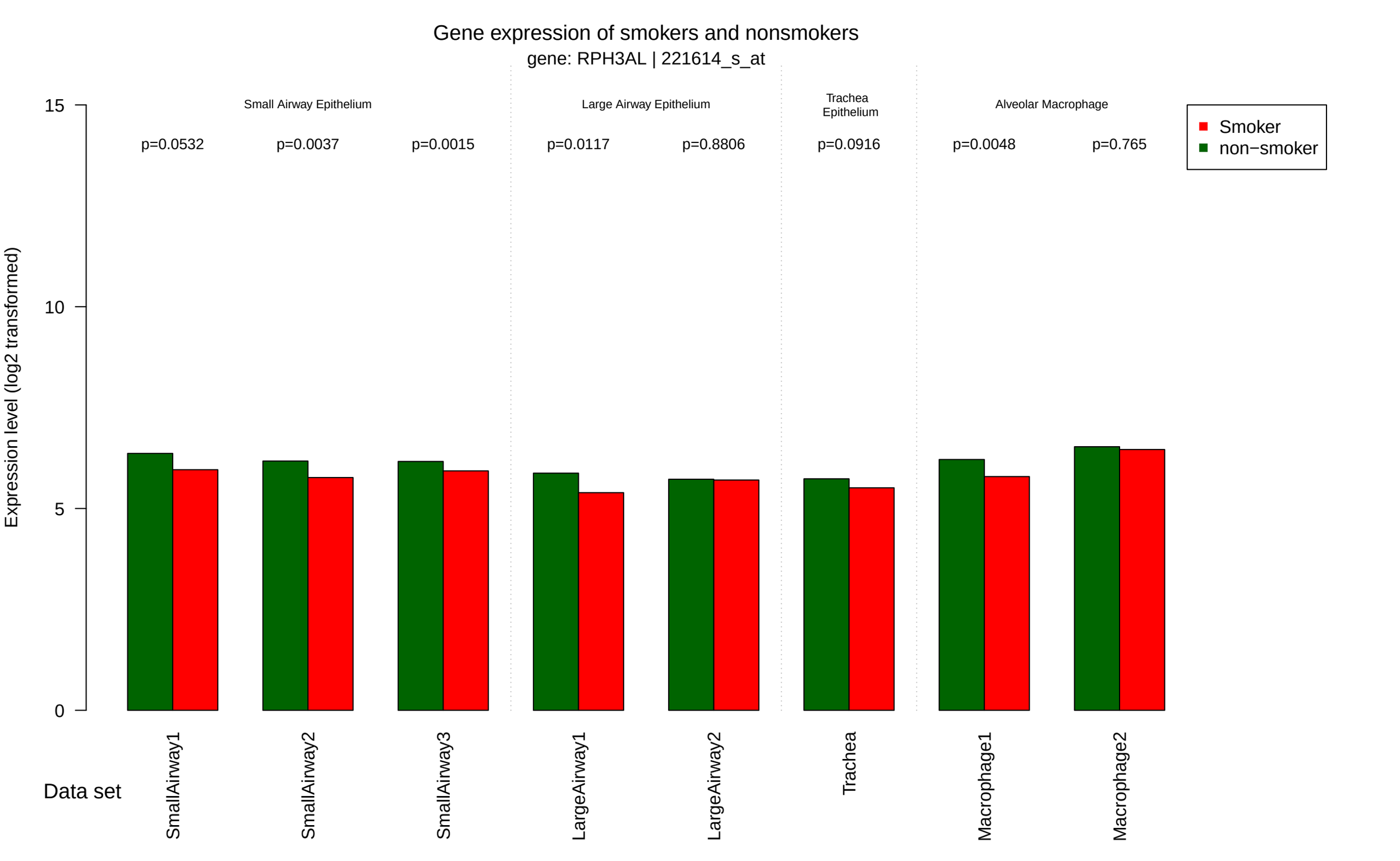
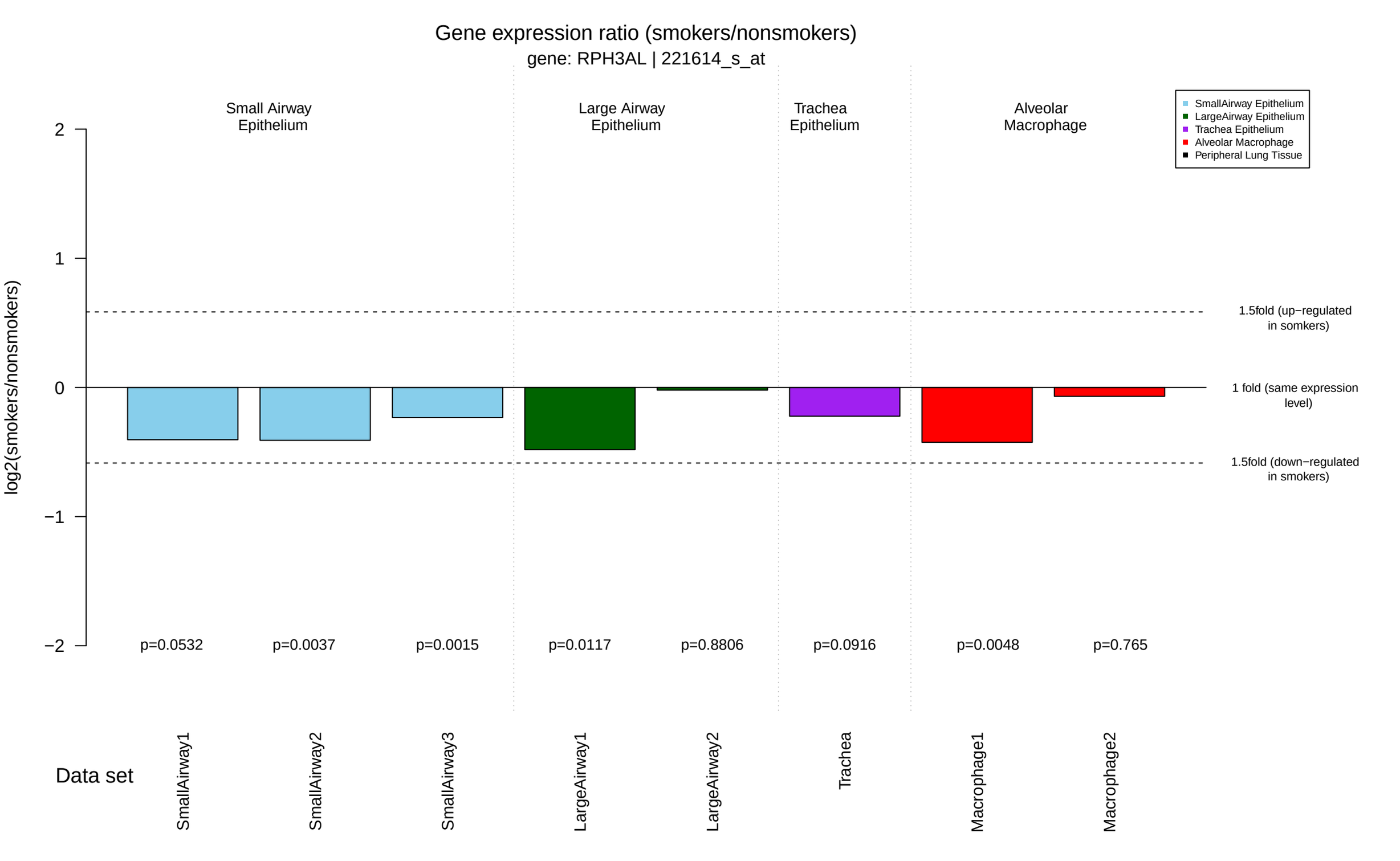
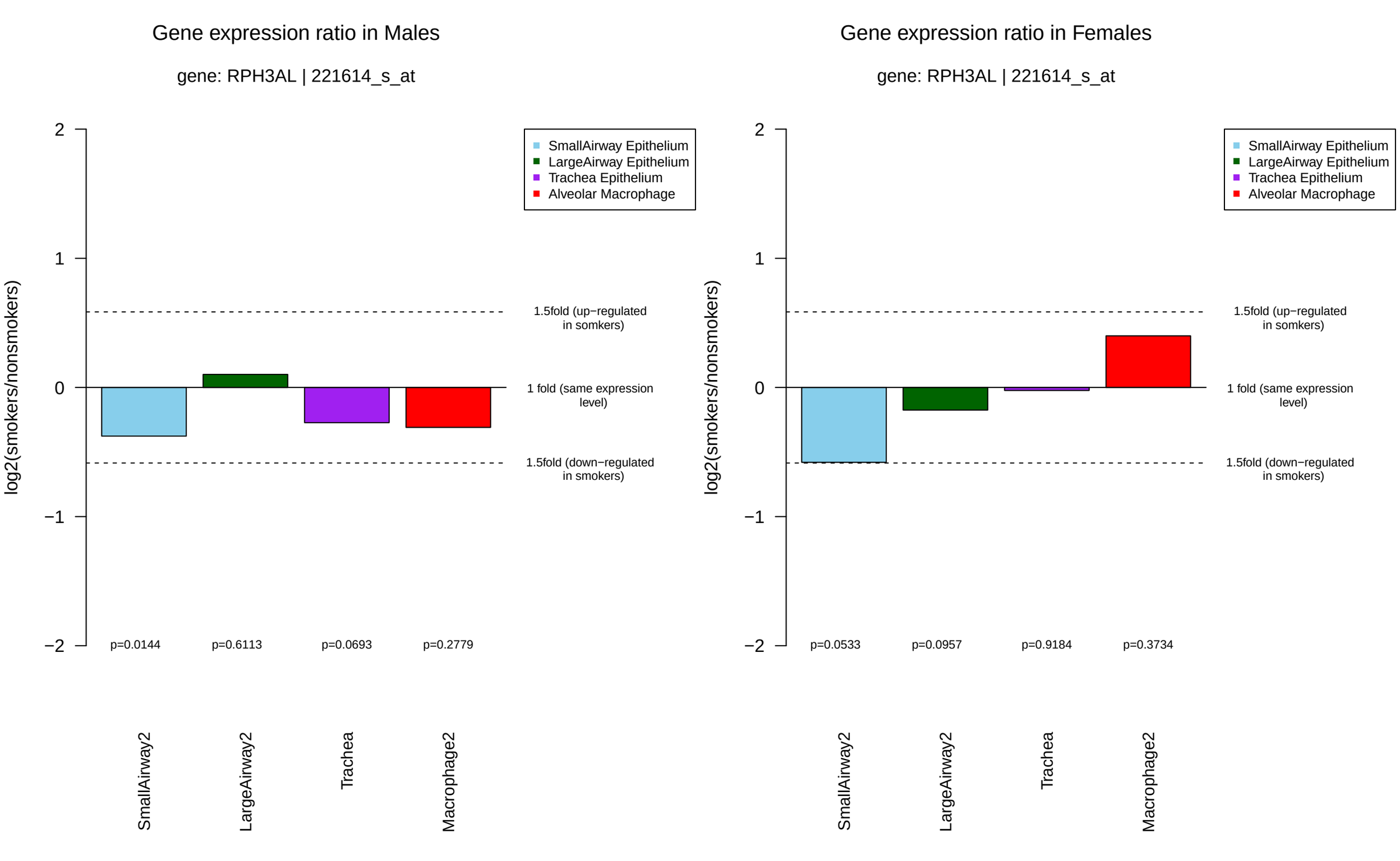
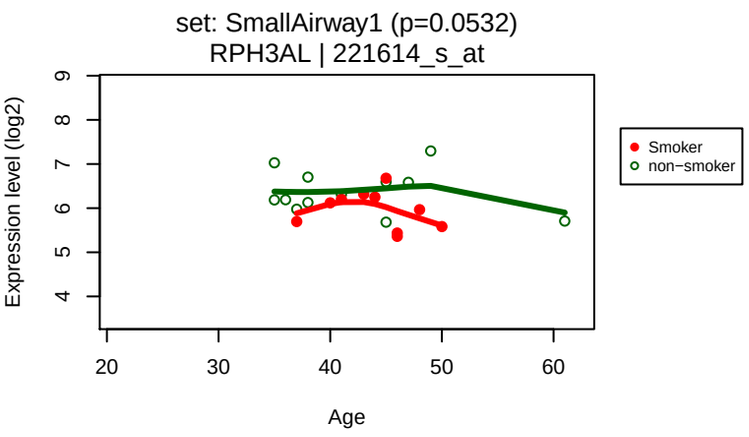
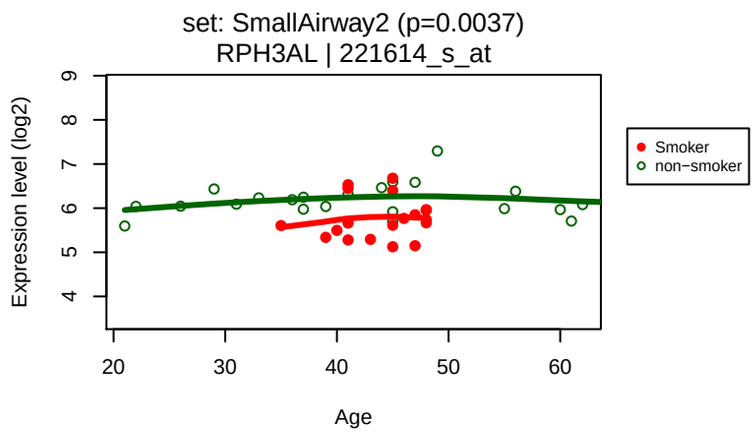
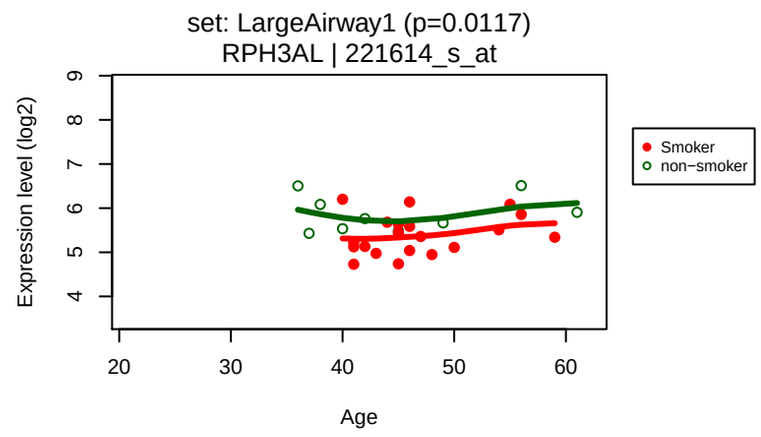

Gene expression of smokers and nonsmokers
gene: RPH3AL | 221614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0532
p=0.0037
p=0.0015
p=0.0117
p=0.8806
p=0.0916
p=0.0048
p=0.765
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

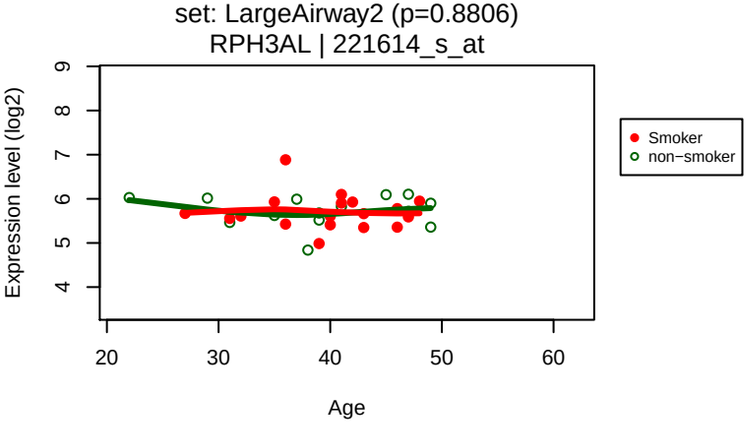

Gene expression ratio (smokers/nonsmokers)
gene: RPH3AL | 221614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0532
p=0.0037
p=0.0015
p=0.0117
p=0.8806
p=0.0916
p=0.0048
p=0.765
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

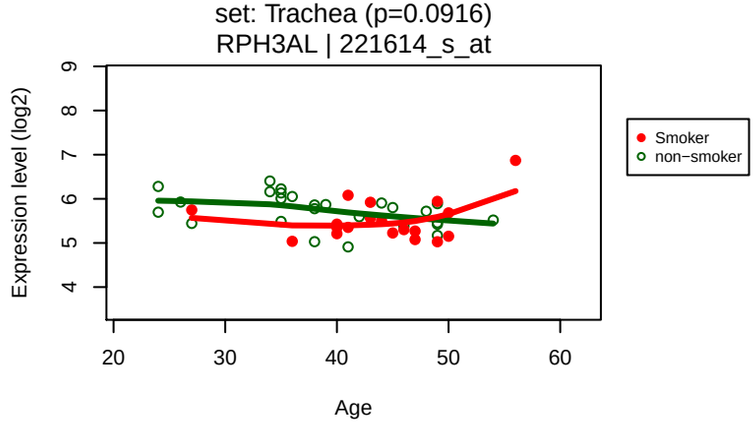

Gene expression ratio in Males
gene: RPH3AL | 221614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0144
p=0.6113
p=0.0693
p=0.2779
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPH3AL | 221614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0533
p=0.0957
p=0.9184
p=0.3734
SmallAirway2
LargeAirway2
Trachea
Macrophage2

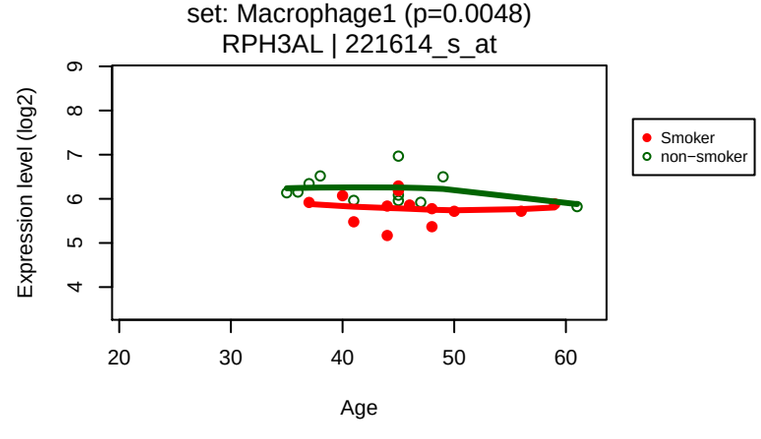

set: SmallAirway1 (p=0.0532)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker

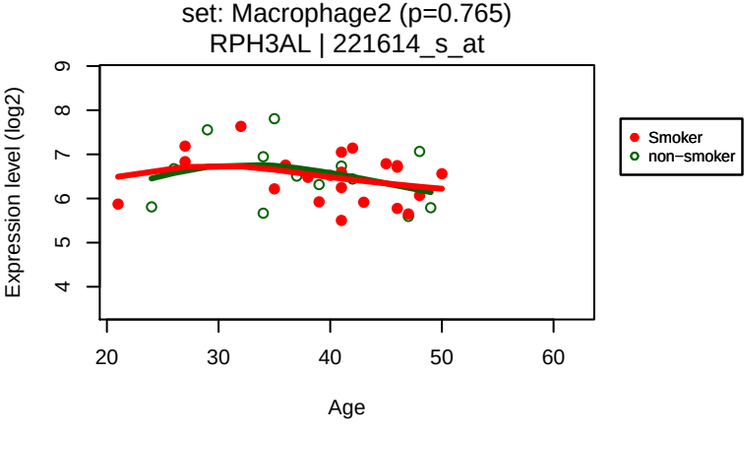

set: SmallAirway2 (p=0.0037)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker

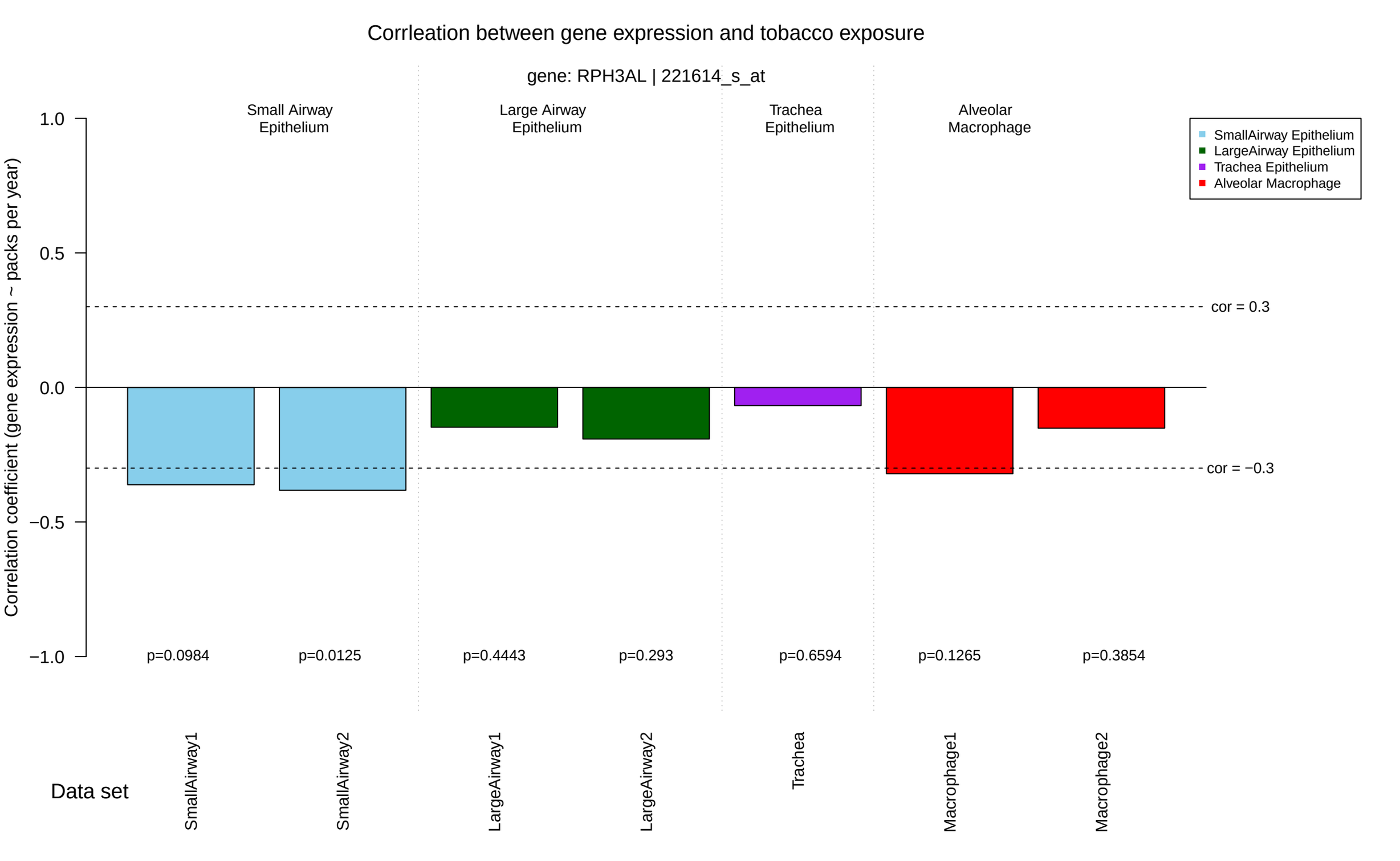

set: LargeAirway1 (p=0.0117)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8806)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0916)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0048)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.765)
RPH3AL | 221614_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPH3AL | 221614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0984
p=0.0125
p=0.4443
p=0.293
p=0.6594
p=0.1265
p=0.3854
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2